

Matthew Faw

+1 (336)-262-1938
mfaw3@gatech.edu
matthewfaw.github.io
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Research Interests

Stochastic Optimization, Online Decision-Making, Reinforcement Learning and Control

Education

- 2024–present **ARC Postdoctoral Fellow**, *Georgia Institute of Technology*, Atlanta, GA.
Mentors: Siva Theja Maguluri, Sahil Singla
- 2018–2024 **Ph.D. ECE (Thesis: Adaptive Algorithms for Stochastic Optimization and Bandit Learning)**, *The University of Texas at Austin*, Austin, TX.
Advisors: Sanjay Shakkottai, Constantine Caramanis.
- 2013–2017 **B.S.E. Electrical & Computer Engineering, B.S. Computer Science, A.B. Math**, *Duke University*, Durham, NC.
Advisors: Nick Buchler, Richard Fair, Benjamin C. Lee

Publications (Google Scholar)

In Submission

- 2026 “Online Allocation with Samples”, **F**, S. T. Maguluri
- 2026 “Online Control with Multiple Sensors”, **F**, S. T. Maguluri

Conference Papers

- ICML 2025 “On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback”, **F**, C. Caramanis, J. Hoffmann
- ICML 2025 “In-Context Fine-Tuning for Time-Series Foundation Models”, **F**, R. Sen, Y. Zhou, A. Das
- COLT 2023 “Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD”, **F**, L. Rout, C. Caramanis, S. Shakkottai
- COLT 2022 “The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance”, **F**, I. Tziotis, C. Caramanis, A. Mokhtari, S. Shakkottai, R. Ward
- SIGMETRICS 2022 “Learning To Maximize Welfare with a Reusable Resource”, **F**, O. Papadigenopoulos, C. Caramanis, S. Shakkottai
- SODA 2022 “Single-Sample Prophet Inequalities via Greedy-Ordered Selection”, C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order)
- NeurIPS 2020 “Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions”, **F**, R. Sen, K. Shanmugam, C. Caramanis, S. Shakkottai

Journal Papers

- Theory of Computing 2026 “Single-Sample Prophet Inequalities via Greedy-Ordered Selection”, C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order)
- POMACS 2022 “Learning To Maximize Welfare with a Reusable Resource”, **F**, O. Papadigenopoulos, C. Caramanis, S. Shakkottai
- TOCS 2017 “Computational Sprinting: Architecture, Dynamics, and Strategies”, S. Zahedi, S. Fan, **F**, E. Cole, B. Lee

Awards + Honors

- 2023 Dr. Brooks Carlton Fowler Endowed Presidential Graduate Fellowship in ECE, 2023-2024 academic year
- 2022 Top 10% reviewer for NeurIPS'22 and AISTATS'22, Highlighted reviewer for ICLR 2022
- 2020 NXP Foundation Fellowship, 2020-2021 academic year

- 2017 Cum Laude Graduation Honors, Duke University
- 2016 Member, Tau Beta Pi and Eta Kappa Nu Honor Societies, Duke University
- 2014 Gold medal, International Genetically Engineered Machine Competition

Talks and Poster Presentations

Invited Talks

- October 2025 INFORMS Job Market Showcase, Atlanta, GA: "Optimal Control with Multiple Sensors"
- July 2025 INFORMS APS Conference, Atlanta, GA: "Fundamental Limits of Regret Minimization in Stochastic Bandits with Evolving Biased Feedback"
- February 2025 Google Research, (Virtual): "Order-Optimal Convergence Rates with Adaptive SGD"
- October 2024 INFORMS Annual Meeting, Seattle, WA: "Order-Optimal Convergence Rates with Adaptive SGD"
- March 2024 Georgia Tech ARC Colloquium, Atlanta, GA: "The Power of Adaptivity in SGD"

Talks

- April 2026 Applied Probability Group Meeting, Atlanta, GA: "Online Control with Multiple Sensors"
- July 2023 COLT 2023, Bangalore, India: "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"
- April 2023 IFML Workshop, UW "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"
- July 2022 COLT 2022, London, UK: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"
- June 2022 SIGMETRICS 2022, IIT Bombay, Mumbai, IN: "Learning To Maximize Welfare with a Reusable Resource"
- April 2022 Machine Learning Lab Research Symposium, UT Austin: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"
- January 2022 SODA 2022, Virtual: "Single Sample Prophet Inequalities via Greedy-Ordered Selection"

Poster Presentations

- July 2025 ICML 2025, Vancouver, Canada, "In-Context Fine-Tuning for Time-Series Foundation Models" and "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"
- May 2024 IFML NSF Site Visit Poster Session, UT Austin, "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"
- December 2023 NeurIPS 2023 Workshop: Algorithmic Fairness through the Lens of Time, New Orleans, LA, "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"
- October 2022 Joint IFML/Data-Driven Decision Processes Workshop, Simons Institute, UC Berkeley, "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"
- December 2020 NeurIPS 2020, Virtual, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"
- November 2019 Texas Wireless Summit, UT Austin, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"

Conference Reviewing

- 2021-Present AISTATS, ALT, ICLR, ICML, JMLR, NeurIPS

Industry Experience

- March–Sept. 2025 **Visiting Researcher**, *Google Research*, Remote.
- June–Sept. 2024 **Ph.D. Student Researcher**, *Google Research*, Mountain View, CA.
- June'17–July'18 **Software Engineer**, *Verato*, McLean, VA.
- May–Aug 2016 **Software Engineering Intern**, *Stateflow Semantics*, *MathWorks*, Natick, MA.

Undergraduate Research Experience

- Jan-Dec 2016 **Datacenter Architecture**, Advisor: Dr. Benjamin Lee, Duke University.
Jan-Dec 2015 **Microfluidics**, Advisor: Dr. Richard Fair, Duke University.
May-Nov 2014 **Synthetic Biology**, Advisor: Dr. Nick Buchler, Duke University.

Teaching Experience

- Spring 2026 **ISyE 3770 Statistics and Applications Instructor**, Georgia Tech.
Served as the instructor for introductory probability and statistics course. Designed and delivered lectures, office hours, homework, and exams for 60 undergraduate students. Managed two undergraduate TAs.
- Fall 2018, **EE 460J, Data Science Lab TA**, UT Austin.
- Spring 2019 Led lab sessions, and graded homeworks and exams
- Spring 2017 **CS 308, Software Design and Implementation TA**, Duke University.
Personally mentored 3 undergraduate CS students, conducted code reviews, advised and evaluated design and implementation of 3 software projects.
- Fall 2015 **ECE 280, Signals & Systems TA**, Duke University.
Held office hours and graded assignments for 70 undergraduate students.
- Spring 2015 **Synthetic Biology House Course Co-Designer/Instructor**, Duke University.
Co-designed and taught the first-offered Duke course on synthetic biology to 10 undergraduates.

Graduate Coursework

- UT Austin Probability & Stochastic Processes, Advanced Probability, Stochastic Processes I, Theoretical Statistics, Online Learning, Large Scale Optimization I & II, Combinatorial Optimization, Sublinear Algorithms, Markov Chains & Mixing Times, Combinatorics & Graph Theory, Analysis & Design of Communication Networks

Technical Skills

- Programming Python, Java, C/C++, JavaScript
Infrastructure Kubernetes, AWS, Google Cloud, Mongo, Solr

References

- Sanjay Shakkottai, UT Austin, sanjay.shakkottai@utexas.edu
Constantine Caramanis, UT Austin, constantine@utexas.edu
Siva Theja Maguluri, Georgia Institute of Technology, siva.theja@gatech.edu